

CLOUD TECH SMART SOLUTIONS

Smart Home System Architecture

System Documentation & Schematics

Project ID:	STR-001
Location:	Struga
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Maintenance Contract:	Active (Standard SLA)
Last Maintenance:	12 May 2026
Next Maintenance Due:	May 2027

This document is restricted to property owners and authorized personnel only.

SECTION 3: SMART HOME TOPOLOGY & DEVICES

3.1 Core System Architecture

The smart building is engineered on the decentralized KNX Bus System to ensure 100% hardware reliability. High-level automations, remote control interfaces, and scheduling profiles are governed by a secure local server running Home Assistant.

3.2 Integrated Accessories & Hardware

The installation features high-end custom user switches from Bticino (Living Now Series) integrated with KNX taster interfaces, and a premium IP door intercom from Vimar Elvox IP (Pixel Series) connected to the local touchscreens.

SECTION 3: SMART HOME TOPOLOGY & DEVICES

KNX Device Registry

Address	Device Type	Channels	System Function	Area / Load
1.1.1	KNX Bus Power Supply 640mA	-	Main Bus Power Injection	Main Switchboard
1.1.2	KNX-IP Interface Gateway	-	Local Network Ethernet Bridge	Network Cabinet
1.1.10	Bticino Relays Actuator	12-Ch Switch	Main Switch Relays (Lights/Outlets)	Ground Floor
1.1.11	Bticino Relays Actuator	12-Ch Switch	Secondary Switch Relays	First Floor
1.1.12	Dimming Actuator	4-Ch Dimmer	Dimmed LED Spotlights Control	Living/Kitchen Rooms
1.1.13	Shutter Actuator	8-Ch Shutter	Blinds & Motorised Curtains Control	Bedrooms / Balcony
1.1.14	Heating Actuator	6-Ch Valve	Underfloor Heating Loop Electrovalves	Climate Loops
1.1.20	KNX Motion / Lux Sensor	4-Ch Sensors	Ceiling Lux/Presence Integration	Hallways / Baths
1.1.30	Bticino Wall Switches	6-Button Taster	Living Now integrated switches	All Living Zones
1.1.40	Vimar Intercom	Pixel IP Outdoor	Video Intercom Doorbell Panel	Main Gate / Entrance

3.3 Topological Data Stream Diagram

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[ Internet / WAN ] <--> [ Secure Wi-Fi Router ] <==> [ Home Assistant Server (Automation Node) ]
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    ||
    || (Local IP LAN Interface)
    \ /
[ KNX IP Gateway ] (Physical Address 1.1.2)
    ||
===== KNX Twisted Pair Bus Line (Green Cable) =====
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|                                     |                                     |                                     |
|                                     |                                     |                                     |
|                                     |                                     |                                     |
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/                                     /                                     /                                     /

[ KNX Power Supply ]      [ Relay Actuators ]      [ Shutter Controller ]      [ Wall Push Buttons ]
(1.1.1)                  (1.1.10 - 1.1.11)          (1.1.13)                               (1.1.30)

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ticino

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